

COMPARISON OF HARD STEP TAKEOVER VS. C^2 BUMPLESS TORQUE BLENDING

Eliminating motor current inrush spikes ($7.4\text{ A} \rightarrow 1.8\text{ A}$) and infinite torque rate-of-change ($\dot{u}$) during human intervention

✗ HARD STEP SWITCHING (DISCONTINUOUS TAKEOVER)

(a) Commanded Torque $u(t)$ – Discontinuous Step

$\Delta u = +0.7\text{ Nm}$ (0 ms)

(b) Torque Rate $\dot{u}(t)$ – Dirac Impulse $\delta(t)$

$\dot{u} \rightarrow \infty$ (Gear Shock)

(c) Phase Current $I(t)$ – 7.4 A Inrush Spike (Overcurrent)

$I \rightarrow 7.4\text{ A}$ (Inrush Spike)

PHYSICAL HARDWARE DAMAGE:

- Inrush current trips inverter overcurrent fault (OCP).
- Instantaneous torque step shears harmonic drive teeth.
- Robot jerks violently against human operator's hand.

✓ C2 BUMPLESS BLENDING (SIGMOID INTERPOLATION)

(a) Commanded Torque $u(t)$ – Smooth Sigmoid S-Curve

$T_{\text{blend}} = 100\text{ ms}$

(b) Torque Rate $\dot{u}(t)$ – Bounded Bell Curve ($\leq 11\text{ Nm/s}$)

$\dot{u}_{\text{max}} = 10.8\text{ Nm/s}$

(c) Phase Current $I(t)$ – Continuous Delivery (1.8 A)

Continuous 1.8 A (Safe)

PHYSICAL SYSTEMS ADVANTAGES:

- Zero inverter overcurrent trip risk; 100% uptime.
- Gearbox torque derivative strictly bounded within mechanical specs.
- Intuitive, smooth haptic feel for the human co-pilot.